IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

SELECTED TOPICS IN POWER SYSTEMS

Wednesday, 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. Question 1 & 2 should be answered in Booklet A, Questions 3 & 4 in Booklet B.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, F. Teng

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Answer ALL Questions (answer Q1 and Q2 in one booklet and Q3 and Q4 in a separate one)

Question 1 on Flexible AC Transmission System (FACTS)

1. A long transmission line interconnects two large AC power systems. The voltages at both ends of the transmission line are maintained at 1 p.u. The reactance of the transmission line is 0.4 p.u. and its resistance and shunt charging can be neglected. The system operator is planning to install shunt reactive compensation at the midpoint of the transmission line to regulate the midpoint voltage at 1 p.u.. For this, the following three options are being considered: i) an SVC with a rated susceptance of 2 p.u. ii) an SVC with a rated susceptance of 3.5 p.u. and iii) a STATCOM with a current rating of 3.2 p.u.

   a) Calculate the increase in steady-state stability limit (with respect to the uncompensated case) that can be achieved with each of the three shunt reactive compensation option stated above. Sketch the power-angle curve for each case and label the salient points.

   [15]

   b) Calculate the critical clearing angle (in degrees) with the SVC having 3.5 p.u. susceptance considering 1.6 p.u. of active power transfer through the line. Assume a self-clearing fault and zero power transfer during the fault.

   [5]

Question 2 on System Operation with Low Inertia

2. a) To maintain the frequency stability in the future low-inertia system, it is important to meet the system frequency requirements on Rate of Change of Frequency (RoCoF), Frequency Nadir and Steady-state Frequency

i) Analytically derive the RoCoF constraint in terms of H_i (inertia constant of individual generators) and RoCoFmax from the swing equation.

[2]

ii) Due to the challenge on measuring the instantaneous RoCoF, the average RoCoF over 500ms has been often used to trigger protection. For the same level of RoCoF limit (e.g. 0.5 Hz/s) and the same largest generation loss, discuss how instantaneous RoCoF and average RoCoF limits would impact the required level of system inertia differently?

[2]

iii) The constraint that guarantees the system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(\frac{H}{f_0} - \text{EFR} \cdot \frac{\text{T}_{\text{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\text{L}} - \text{EFR})^2 \cdot \text{T}_{\text{d}}}{4 \cdot \Delta f_{\max}}$$

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_{L}: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
$\Delta f_{\max}$: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
T_{d}: Delivery-time of frequency response (Units: s).
T_{EFR}: Delivery-time of EFR (Units: s).

Other assumptions:

1. The largest generation unit is 1.8 GW and with reliability of 99%
2. Other numerical values: f_0 = 50 Hz, Δf_{max} = 0.8 Hz, T_d = 10 s, T_{EFR} = 0.5 s, EFR = 200 MW

The forecasted system inertia is following the distribution below:

Inertia (H)	Probability
140 GWs	5%
150 GWs	20%
160 GWs	50%

170 GWs	20%
180 GWs	5%

Draw the probability distribution of not meeting the frequency nadir requirement at various system inertia levels, assuming the frequency response is calculated and scheduled based on such level of inertia. To maintain the probability of meeting the frequency nadir requirement above 99.9%, calculate how much frequency response (R) the system operator needs to procure.

[4]

iv) In addition to frequency response (R) with delivery time T_d and enhanced frequency response (EFR) with delivery time T_{EFR}, assume there is a new service (INS) that can be delivered instantly. Update the system frequency nadir requirement in (iii) to include this new service

[2]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) A wind turbine with a power rating of 12 MW operates above rated wind speed with a deloading of 1 MW (ie it generates 11 MW). Calculate the highest synthetic inertia constant, *H*, the wind turbine could emulate under this operating point if the worst RoCoF expected is of -2 Hz/s and the system's nominal frequency is 50 Hz.

[3]

ii) Which of the following forms of frequency support service from renewable generation would be more effective at reducing the worst RoCoF at the beginning of a frequency excursion caused by the loss of a large generator unit: synthetic inertia or automatic generation control (AGC) droop emulation? Justify your answer

[4]

iii) A wind turbine manufacturer suggests that their wind turbine would be able to boost its output power by up to 0.3 pu for a short period of time (e.g. 10 s) when operating at point A of its power curve (see Figure 2.1). Is it possible? Justify your answer.

[3]

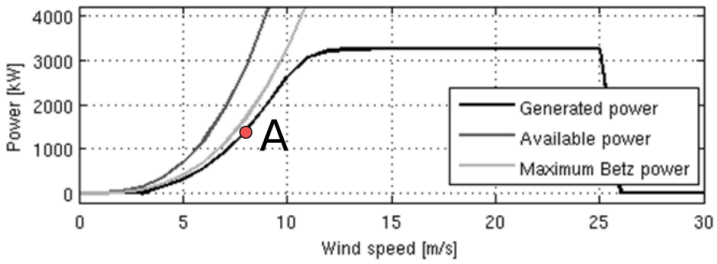


Figure 2.1 – Power curve of a 3.6 MW wind turbine.

Question 3 on Power System State Estimation

(i) Why do system operators need a state estimator? [3]

(ii) What are the fundamental differences between state estimation in transmission and distribution network operation? [5]

(iii) In the context of power transmission state estimation starting with measurements and states, formulate weighted least square (WLS) as maximum likelihood estimate (MLE). [5]

(iv) What is equality constrained state estimation? [3]

(v) Under what circumstances constrained state estimation is adopted? [2]

(vi) What is the role of critical measurement in state estimation? [2]

Question 4 on Integrated Energy Systems

Considering the electrification of heat sector, heat appliances through thermal energy storage (TES) could provide flexibility to the electricity system and support more cost-effective integration of variable wind generation. Consider a system with capacity of wind generation of 35GW, nuclear generation 12.4GW, with the rest of the load being supplied by CCGT plants, which are all the same type with same characteristics. Electricity demand includes non-heat demand and electrified heat demand. Operating reserve is required to deal with uncertainties in wind generation production, which is managed by part-loading CCGT.

Key characteristics of the CCGT plant are:
Capacity: 600 MW
Minimum Stable Generation (MSG): 300 MW
Minimum online time: 2 hours (once a CCGT unit is started up, it would be running for at least 2 hours)

System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

	1st hour	**2nd hour**
Wind available (MW)	14,600	28,800
Nuclear (MW)	12,400	12,400
Electricity Demand (MW) - Non-heat demand - Electrified heat demand	51,000 36,000 15,000	51,000 36,000 15,000
Reserve requirement (MW)	9,600	13,500

a) Assuming that
   - TES is not available,
   - 6 CCGT units have already been online for more than 1 hour at the beginning of the 1st hour,

   answer the following questions:

   i) how many extra CCGT units should be started up in the 1st hour?
   [4.5]

   ii) Calculate wind curtailment in the 2nd hour.
   [4.5]

b) TES can accommodate surplus of wind power and supply heat demand.
   Assume that
   - 3500MW (power equivalent) TES is available,
   - The initial thermal energy stored in TES is 3000MWh (power equivalent),
   - Heat loss of TES is ignored.
   - 6 CCGT units have already been online for more than 1 hour at the beginning of the 1st hour,

   and answer the following questions:

   i) calculate the minimum wind curtailment in the 2nd hour.

[6]

ii) If CO_2 emission associated with the operation of CCGT is 0.4ton/MWh, how much CO_2 is saved by using TES?

[5]